AITS
ALL INDIA TEST SERIES

Dropper JEE Advanced

AJPA11.1/10 **Test- 10**

DURATION : 180 Minutes **DATE : 16/03/2025** **M.MARKS : 198**

Topics covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 54 questions. The maximum marks are **198**.

SECTION 1 (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

SECTION 2 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is(are) chosen;

Partial Marks : + 3 If all the four options are correct but ONLY three options are chosen;

Partial Marks : + 2 If three or more options are correct but ONLY two options are chosen, both of which are correct;

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : –2 In all other cases.

SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered;

Zero Marks : 0 In all other cases.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Students (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^{2}kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

**** Use above values unless otherwise specified in a question.***

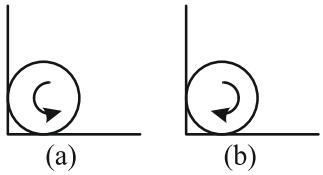
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PART-I (PHYSICS)

SECTION-1

Single Correct Type Questions:

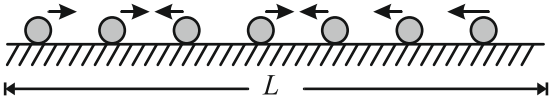
1. A sphere is placed rotating with its centre initially at rest in a corner as shown in figure (a) & (b). Coefficient of friction between all surfaces and the sphere is $\frac{1}{3}$. Find the ratio of the frictional force $\frac{f_a}{f_b}$ by ground in situations (a) & (b)



(A) 1 (B) $\frac{9}{10}$

(C) $\frac{10}{9}$ (D) $\frac{7}{9}$

2. There are n identical balls travelling along a straight line on a smooth horizontal table. All balls have equal speed v m/s and each one is travelling to either right or left. All collisions between the balls are head on elastic. At some point in time all balls will have fallen off the table. The time at which this happens will definitely depend on initial positions of the balls. Over all possible initial positions of the balls, what is the longest amount of time that you would need to wait to ensure that the table has no more balls left on it?



(A) $\frac{nL}{v}$ (B) $\frac{L}{nv}$

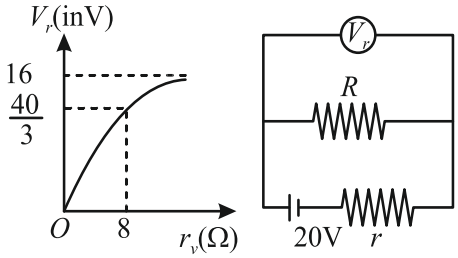
(C) $\frac{2L}{v}$ (D) $\frac{L}{v}$

3. Consider a solid insulating sphere (of relative permittivity one) of radius R having a spherically symmetric charge distribution. The charge density varies as $\rho = \rho_0 r^2$ (ρ_0 is a constant and r is measured from centre). Consider two points A and B at distances x and y respectively ($x < R$, $y > R$) from the centre. If magnitude of electric fields at points A and B are equal, then

(A) $x^2y = R^3$ (B) $x^3y^2 = R^5$

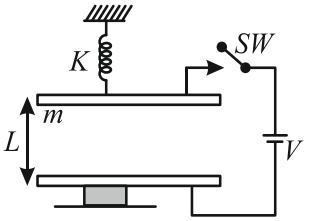
(C) $x^2y^3 = R^5$ (D) $x^4y = R^5$

4. A real cell of emf 20 V is connected across a resistor of resistance R. A voltmeter of varying resistance (r_v) is connected across resistance to measure the potential drop against r_v as shown in figure. Thus, value of R will be (in ohm)



(A) 2
(B) 6
(C) 8
(D) 4

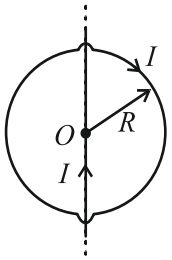
5. An uncharged parallel plate capacitor having its lower end fixed and upper end is attached with spring having spring constant K. Upper plate is in equilibrium before switch (SW) is closed. After switch is closed, the maximum value of the potential of battery (V) so that the system can acquire new equilibrium position is. (mass of upper plate = m, area of plate = A)



(A) $\sqrt{\left(\frac{5}{3}\right)^3 \frac{L^3K}{\varepsilon_0 A}}$ (B) $\sqrt{\left(\frac{2}{3}\right)^3 \frac{L^3K}{\varepsilon_0 A}}$

(C) $\sqrt{\left(\frac{5}{4}\right)^2 \frac{L^3K}{\varepsilon_0 A}}$ (D) $\sqrt{\left(\frac{3}{4}\right)^2 \frac{L^3K}{\varepsilon_0 A}}$

6. A circular conducting loop of radius R carries a current I. Another straight infinite conductor carrying current I passes through the diameter of this loop as shown in the figure. The magnitude of force exerted by the straight conductor on the loop is:



(A) $\pi\mu_0 I^2$

(B) $\mu_0 I^2$

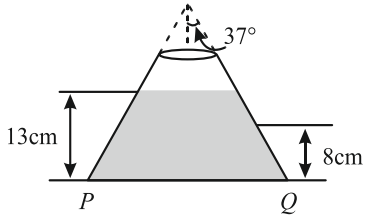
(C) $\frac{\mu_0 I^2}{2\pi}$

(D) 0

SECTION-2

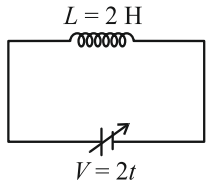
One or More Than One Correct Type Questions:

7. A conical shaped vessel has semi vertical angle 37°. The vessel is filled with water (density ρ = 1000 kg/m^3) upto a height of 13 cm and placed on a smooth horizontal surface. Upper surface is opened to atmosphere. A hole of area of cross-section 1.5 cm^2 is made on curved wall at a height of 8 cm from bottom as shown in figure. Area of water surface in vessel is much large as compared to the area of the hole. The vessel is held in equilibrium by applying a suitable horizontal force Select the correct option(s).
(Take g = 10 m/s^2)



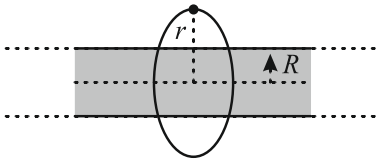
(A) Initial velocity of efflux is 1 m/s
(B) Initial horizontal range of water jet from point Q is 7.25 cm
(C) Horizontal force required to keep the vessel in static equilibrium is 0.15 N
(D) Horizontal force required to keep the vessel in static equilibrium is 0.12 N

8. A time varying voltage $V = 2t$ (volt) is applied across an ideal inductor of inductance L = 2 H. At t = 0, current is zero. Select the correct option(s)?



(A) Current flowing through inductor versus time graph is a parabola
(B) Energy stored in the inductor in the form of magnetic field at t = 2 s is 4 J
(C) Energy stored in the inductor at time t = 1 s is increasing at a rate of 1 J/s
(D) Energy stored in the inductor in the form of magnetic field is constant all the time

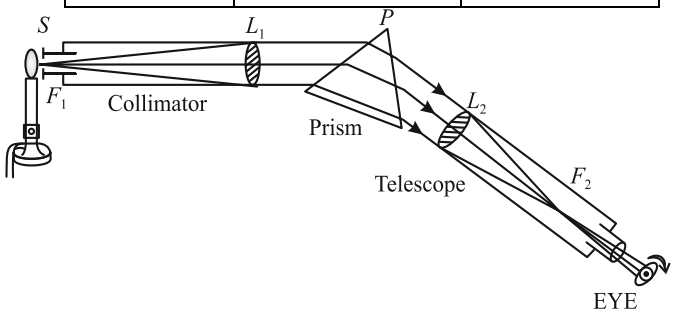
9. Consider a long hypothetical cylindrical hollow planet of radius R and a fixed uniform mass per unit area σ. If a satellite revolves around the planet in circular orbit of radius r in a plane perpendicular to the axis of the planet, then:-



(A) Kinetic energy of satellite is independent of orbital radius of satellite.
(B) Kinetic energy of satellite depends on radius of planet.
(C) Orbital speed of satellite depends on orbital radius of satellite.
(D) Time period revolution of satellite is directly proportional to its orbital radius.

10. The common prism spectroscope is shown in figure is generally made up of three parts, viz., a prism and two telescopes, One of these telescopes is used to render the rays which fall on the prism parallel to teach other, and is therefore called the collimator, the other telescope is made movable so that it can be turned into the proper direction for observing any desired colour, and is, therefore, called the view-telescope. A circular scale is also provided to measure angular position of telescope. On the circular scale 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. The smallest division of the main scale is half-a-degree (0.5°). Initially view-telescope is focused to collect rays directly from collimator and position of telescope is measured. This is position (1). Now a prism is placed as shown in the figure and telescope is focused for red, yellow and violet. The respective positions of telescopes are (2), (3) and (4).

	Reading of main scale	Reading of veriner scale
Position (1)	20.0°	20
Position (2)	30.0°	16
Position (3)	32.5°	9
Position (4)	33.5°	11



The deviation for:
(A) red light is 10°4'
(B) violet light is 13°19'
(C) yellow light is 12°19'
(D) violet light is 13°21'

11. A Carnot heat engine draws heat from a reservoir at temperature T_1 and rejects heat to another reservoir at temperature T_3. The Carnot forward cycle engine drives a Carnot reversed cycle engine or Carnot refrigerator which absorbs heat from reservoir at temperatures T_2 and rejects heat to a reservoir at temperature T_3. The high temperature T_1 is 600 K and low temperature T_2 is 300 K.

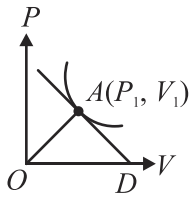
(A) If heat supplied to engine Q_1 is equal to heat absorbed by the refrigerator Q_2, the value of T_3 is 400 K

(B) The C.O.P (coefficient of performance) of Carnot refrigerator is 2

(C) The C.O.P (coefficient of performance) of Carnot refrigerator is 3

(D) If heat supplied to engine Q_1 is equal to heat absorbed by the refrigerator Q_2, the value of T_3 is 450 K

12. n moles of an ideal gas undergo an isothermal process at temperature T. P-V graph of the process is as shown in the figure. A point A (P_1 V_1) is located on the P-V curve. Tangent at point A, cuts the V-axis at point D. AO is the line joining the A to the origin O of PV diagram. Then,



(A) coordinates of points D is $\left(\frac{3V_1}{2}, 0\right)$

(B) coordinates of points D is $(2V_1, 0)$

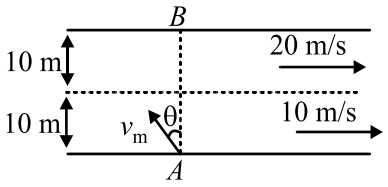
(C) area of the triangle AOD is nRT

(D) area of the triangle AOD $\frac{3}{4}nRT$

SECTION-3

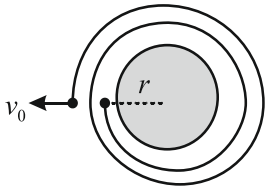
Numerical Value Type Questions:

13. In a river of 20 m width. Half part of river flows with speed 10 m/s and remaining half part flows with 20 m/s as shown in figure. A man starts to swim from A and lands at point B on the opposite bank in 10 s. Man swims with speed v_m with respect to river at an angle θ with line AB. Man swims with constant speed and in same direction with respect to river throughout the motion. Then, the value of $\frac{2}{3}\tan\theta$ is



14. During a thrilling football match, a player kicks a football from the ground towards the goal post, which is 50 meters away. The ball majestically sails through the air, clearing the goal post at a height of 2 meters above the top of the post, and continues to fly, finally landing 10 meters behind the goal post. Remarkably, as the ball crosses above the goal post, its speed is precisely 20 meters per second. Assuming the ball follows a perfect parabolic trajectory and neglecting air resistance. Take g = 10 m/s^2. If the height of the goal post is h (in m), the value of h is

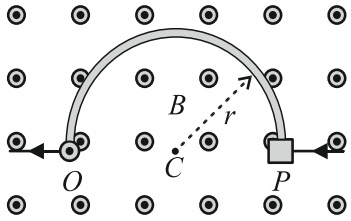
15. A cylinder of radius r = 7 m fixed on a smooth horizontal surface with its axis vertical. One end of an ideal string of length L = 140 m is attached to the surface of cylinder close to horizontal surface & is tightly wound on the cylinder and particle of mass m is attached to other end of the string. The particle is on the horizontal surface & is given a horizontal velocity v_0 = 25 m/s directed along the radius vector drawn from the cylinder's axis. Then find the minimum time (in second) after which the string will be fully wound around the cylinder again. Take $\pi = \frac{22}{7}$



16. A protonium atom consists of a proton and an antiproton. An antiproton is negatively charged particle, mass and modulus of charge of which are equal to those of a proton. If binding energy of an electron in hydrogen atom is $\frac{40}{3}$ eV and mass of proton is 1800 times the mass of electron, find the energy of a photon emitted in transition of a protonium atom from the first excited state to ground state. Give your answer in multiples of 'keV'.

17. A sound detector is moving with a constant velocity towards a stationary sound source. The source emits a beep of sound of frequency v_0 = 264 Hz that propagates with speed c = 330 m/s. The detector detects the beep of sound over a distance Δx = 66 m and registers a frequency v = 275 Hz. Find duration of the beep emitted by the source. (in sec)

18. A schematic model of a current limiter consists of a conducting half ring of radius r = 10 cm and mass m = 100 gram, one end O of which is hinged while the other rests on a fixed metallic support P. The ring can rotate in a vertical plane about O without friction in presence of a uniform horizontal magnetic field of induction B = 2 T pointing perpendicular to the plane of the ring. The ring is connected in series in a branch of a circuit through the hinge O and the metallic contact P. Find the minimum current (in ampere), at which the limiter opens the circuit. (Take the acceleration of free fall, $g = 10\ m/s^2$) (C is the center)



PART-II (CHEMISTRY)

SECTION-1

Single Correct Type Questions:

19. Which of the following is the correct statement?

(A) $T\ell^{+3}$ and Al^{+3} act as an oxidizing agent in aqueous solution

(B) In $T\ell I_3$, $T\ell$ is present in +3 and I is present in $-\frac{1}{3}$ oxidisation state respectively.

(C) Hydrogen bonding in H_3BO_3 gives a layered structure

(D) $B(OC_2H_5)_3$ burns with blue edge flame

20. An aqueous solution containing a non-volatile solute is heated in a open vessel till the boiling point is reached. Thereafter, the boiling is continued and the temperature and vapour pressure of solution are recorded at 5 min, 10 min and 15 min. The corresponding temperatures and pressure are T_5, T_{10}, T_{15} and P_5, P_{10}, P_{15} respectively. The correct relationship among them is :

(A) $T_5 < T_{10} < T_{15}$ and $P_5 = P_{10} = P_{15}$

(B) $T_5 < T_{10} < T_{15}$ and $P_5 < P_{10} < P_{15}$

(C) $T_5 = T_{10} = T_{15}$ and $P_5 < P_{10} < P_{15}$

(D) $T_5 = T_{10} = T_{15}$ and $P_5 = P_{10} = P_{15}$

21. Which of the following statements is **INCORRECT** with respect to Boron for its compounds?

(A) Titration of boric acid with NaOH does not give a sharp end point but when taken along with sugars, it gives a sharp end point with NaOH.

(B) Aqueous solution of Borax shows buffer properties.

(C) B_2H_6 on reaction with excess NH_3 at low temperature forms an ionic compound which on heating gives an aromatic compound which is less reactive than benzene.

(D) On hydrolysis of BF_3, boric acid along with fluoroborates are obtained.

22. Select the incorrect statement from the following

(A) The translational kinetic energy of 15 g of ethane is $\frac{3RT}{4}$

(B) The volume of acetylene required for producing 78 g benzene is $\frac{3RT}{P}$, at given temperature T.

(C) Pressure correction term in vander Wall's equation for n moles of gas is $\frac{an^2}{V^2}$.

(D) Hydrogen will diffuse. 3 times faster than ammonia gas at same pressure and temperature.

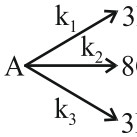
23. Which of the following reactions does not occur in Bessemer converter in the extraction of copper from chalcopyrite?

(A) $2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2$

(B) $FeO + SiO_2 \rightarrow FeSiO_3$

(C) $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$

(D) $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$

24. 

; At time t = 0, initial mole of A is

1. Overall half-life of the reaction is 15 days. Then calculate the number of moles of C after 45 days if the ratio of $k_1 : k_2 : k_3$ is 4: 2 : 1

(A) 1 (B) 2
(C) 3 (D) 4

SECTION-2

One or More Than One Correct Type Questions:

25. Mark out the correct statement(s).
When ethanol, (CH_3CH_2OH) is treated with
(A) Conc. H_2SO_4 at 170°C, it gives ethene $CH_2 = CH_2$.
(B) Conc. H_2SO_4 at 140°C, it gives ether $CH_3CH_2OCH_2CH_3$.
(C) $H_2{}^{18}O$, H_2SO_4 (dil. H_2SO_4) at 140°C it gives ethene.
(D) $H_2{}^{18}O$, H_2SO_4 (dil. H_2SO_4) at 140°C, it gives $CH_3 - CH_2 - {}^{18}OH$

26. Which of the following pairs cannot be separated by concentrated aqueous sodium hydroxide solution?
(A) Al^{+3} and Sn^{+2} (B) Al^{+3} and Fe^{+3}
(C) Al^{+3} and Zn^{+2} (D) Zn^{+2} and Pb^{+2}

27. Which of the following is/are correctly ordered?
(A) $-OH > -NH_2$ (+M effect)
(B) $-SO_3H > -NO_2$ (–M effect)
(C) $-CR_3 > -CHR_2$ (+I effect)
(D) $-CN > -F$ (–I effect)

28. Consider 0.1 molal aqueous solutions of two solutes X and Y. The solute X behaves as a univalent electrolyte, while the solute Y dimerizes in solution. Which of the following statement(s) is/are correct regarding these solutions?
(A) The boiling point of the solution of X will be higher than that of Y.
(B) The osmotic pressure of the solution of X will be higher than that of Y.
(C) The freezing point of the solution of X will be higher than that of Y.
(D) The relative lowering of vapour pressure of both the solution will be the same.

29. The electron affinity of nitrogen is very low which can be attributed to
(A) Nitrogen atom has stable half-filled $2p^3$ configuration
(B) Addition of electron to 2p orbital creates spin paired electron repulsion
(C) Energy released due to attraction between the nucleus and incoming electron is off set by the inter electron repulsion by pairing
(D) Due to small size also, the inter electronic repulsions are more than the other elements in the group

30. The correct statements about peptides are
(A) A dipeptide has one peptide link between two amino acids
(B) By convention, N-Terminus is kept at left and C-terminus at right in the structure of a peptide
(C) α-D-glucopyranose and β-D-glucopyranose are anomers.
(D) Protein and nylon–6, 6 are polyamides.

SECTION-3

Numerical Value Type Questions:

31. Consider the following combination of atomic orbitals
Combining orbitals (Internuclear axis)
(i) $s + p_x(x)$ (ii) $d_{xy} + d_{xy}(x)$
(iii) $d_{yz} + p_z(z)$ (iv) $s + s(z)$
(v) $d_{yz} + d_{yz}(x)$ (vi) $p_y + p_y(y)$
(vii) $d_{z^2} + d_{z^2}(z)$ (viii) $d_{xy} + d_{xy}(z)$
(ix) $p_x + p_x(y)$ (x) $s + p_z(x)$
Then calculate value of $a^2 + b^2 + 2cd$. (where a = no. of σ-bond; b = no. of π bond; c = no. of δ; d = non-bonding combinations)

32. A solution containing 0.012 kg of $Ba(NO_3)_2$ in 0.1 kg of water boils at 100.57°C. Calculate the degree of ionization of the salt. K_b (water) = 0.52 K kg mol^{-1} (Atomic mass of barium = 137.3, N = 14, O = 16 g/mol) (Give answer upto two decimal places)

33. 0.5 mol of an ideal gas 'A' is present in a chamber and 0.5 mol an ideal gas 'B' is present in an other identical chamber. Both the chambers are connected through a valve. Both the gases are at 1 atm pressure and same temperature. What will be the total change in entropy (in J/K) when valve is opened between the chambers? (Given, R = 8.314 J/mol-K, ln2 = 0.693) (Give answer upto two decimal places)

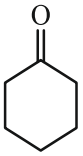
34. The observed rotation of 2 gram of a compound in 10 ml of solution in polarimeter tube of 25 cm long is +24.6°. The specific rotation of the compound (in degree) is

35. The relative reactivity of 1° : 2° : 3° hydrogen to bromination is 1 : 82 : 1600. In the following reaction $CH_3-CH_2-CH_3 \xrightarrow[h\nu]{Br_2}$

$$H_3C-\underset{\underset{Br}{|}}{CH}-CH_3 + CH_3CH_2CH_2Br$$

the percentage yield of 2-Bromopropane is: (Give answer upto two decimal places)

36. In the following reaction



$$+Ph_3\overset{+}{P}-\overline{C}H_2 \rightarrow \underset{\text{(Product)}}{R} + Ph_3PO$$

The percentage mass of hydrogen in the product (R) is:

PART-III (MATHEMATICS)

SECTION-1

Single Correct Type Questions:

37. From a moving point P lying on the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=p$, perpendiculars PA, PB, PC are drawn to the three coordinate planes. The locus of the foot of perpendicular drawn from origin to plane passing through points A, B and C is

(A) $\left(x^2+y^2+z^2\right)\left(\frac{1}{ax}+\frac{1}{by}+\frac{1}{cz}\right)=p$

(B) $\left(x^2+y^2+z^2\right)\left(\frac{1}{ax}+\frac{1}{by}+\frac{1}{cz}\right)=2p$

(C) $\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)\left(\frac{x}{a}+\frac{y}{b}+\frac{z}{c}\right)=p$

(D) $\left(x^2+y^2+z^2\right)\left(\frac{a}{x}+\frac{b}{y}+\frac{c}{z}\right)=p$

38. $\int \sec^{q/3}x \cdot \operatorname{cosec}^{p/3}x\,dx$ equals, where

$(p,q \in I^+ \text{ and } p+q=12)$

(A) $\frac{\tan^{\frac{-p}{3}+1}x}{\frac{-p}{3}+1}+c$

(B) $\frac{\tan^{\frac{-p}{3}+1}x}{\frac{-p}{3}+1}+\frac{\tan^{\frac{-p}{3}+3}x}{\frac{-p}{3}+3}+c$

(C) $\frac{\tan^{\frac{-p}{3}+3}x}{\frac{-p}{3}+3}+c$

(D) $\frac{\tan^{\frac{-p}{3}+1}x}{\frac{-p}{3}+1}-\frac{\tan^{\frac{-p}{3}+3}x}{\frac{-p}{3}+3}+c$

39. Conditions so that equation $(a_1x^2 + b_1x + c_1)(a_2x^2 + b_2x + c_2) = 0$, $(a_1a_2 > 0)$ have four real roots if

(i) $c_1c_2 > 0$

(ii) $a_1c_2 < 0$

(iii) $a_2c_1 < 0$

(A) only (i) is sufficient

(B) either of two statements are sufficient

(C) either of (ii) or (iii) is individually sufficient

(D) all the three statements must be correct

40. $ABCD$ is a cyclic quadrilateral with $AC \perp BD$ and O is the centre of its circumcircle, then $\overrightarrow{OA}\cdot\overrightarrow{OB}+\overrightarrow{OB}\cdot\overrightarrow{OC}+\overrightarrow{OC}\cdot\overrightarrow{OD}+\overrightarrow{OD}\cdot\overrightarrow{OA}$ is equal to

(A) 1

(B) –1

(C) 0

(D) 2

41. If $P_n = 1 + \cos(2^{n-1}\theta) - \sin(2^{n-1}\theta)$ and $Q_n = 1 + \tan 2^{n-1}\theta$ such that $(P_1Q_0)(P_2Q_1)\ldots(P_nQ_{n-1}) = 1$, then $\sin 2^n\theta$ equals

(A) $\cos\theta$

(B) $\sin\theta$

(C) $\cos 2\theta$

(D) $\sin 2\theta$

42. Value of $\sum_{r=1}^{10} {}^{15}C_r \cdot {}^{35}C_{10-r}$ is

(A) ${}^{50}C_{10} - {}^{15}C_{10}$ (B) ${}^{50}C_{10} - {}^{35}C_{10}$

(C) ${}^{40}C_{10}$ (D) ${}^{50}C_{10} - 1$

SECTION-2

One or More Than One Correct Type Questions:

43. Let $\Delta_1 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} bc-a^2 & ac-b^2 & ab-c^2 \\ ac-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ac-b^2 \end{vmatrix}$

and $\Delta_3 = \begin{vmatrix} a & -b & c \\ -b & c & -a \\ c & -a & b \end{vmatrix}$, then choose the correct option(s)

(A) $\Delta_2 = -\Delta_3^2$ (B) $\Delta_2 = \Delta_1^2$

(C) $\Delta_3 = \Delta_1$ (D) $\Delta_3 = -\Delta_1$

44. A six digit number is formed randomly using digits {1, 2, 3} with repetitions. Choose the correct option(s)

(A) Probability that all digits are used at least once is $\frac{20}{27}$

(B) Probability that digit 1 is used odd number of times and 2 is used even number of times is $\frac{182}{3^6}$

(C) Probability that all digits are used as well as odd digits are used odd number of times and even digit is used even number of times is $\frac{50}{3^5}$

(D) Probability that exactly two digits are used is $\frac{62}{243}$

45. Let $n = 2^3 3^4 7^2 11^5$. Choose the correct option(s)

(A) The number of divisors of n, which are of the form $(4k + 2)$, is 89 ($k \in N$)

(B) The number of divisors of n, which are perfect squares is 36

(C) The number of divisors of n^2, which are not divisors of n, is 3105

(D) The number of divisors of n^2, which are not divisors of n is 360

46. $2\tan\left(\tan^{-1}(x) + \tan^{-1}(x^3)\right)$, where $x \in R - \{-1, 1\}$, is equal to:

(A) $\frac{2x}{1-x^2}$

(B) $\tan\left(2\tan^{-1} x\right)$

(C) $\tan\left(\cot^{-1}(-x) - \cot^{-1}(x)\right)$

(D) $\tan\left(2\cot^{-1} x\right)$

47. Let $f(x) = e^{-\frac{1}{x^2}} + \int_0^{\frac{\pi}{2}} \sqrt{1+\sin t}\,dt \; \forall x \in (0,\infty)$ then-

(A) f' exists and is continuous $\forall\ x \in (0, \infty)$

(B) f'' exists $\forall x \in (0, \infty)$

(C) f' is bounded

(D) There exists $\alpha > 0$ such that $|f(x)| > f'(x) \forall x \in (\alpha, \infty)$

48. The vertices A, B, C of ΔABC are 0, 3 and $4i$ on the argand plane. If three equilateral triangles (ΔPBC, ΔQAC and ΔRBA) are constructed outwardly to given ΔABC, then

(A) The distance between centroid of ΔPQR and orthocentre of ΔABC is $\frac{5}{3}$

(B) The distance between centroid of ΔPQR and orthocentre of ΔABC is $\frac{5}{6}$

(C) The distance between centroid of ΔPQR and circumcentre of ΔABC is $\frac{5}{3}$

(D) The distance between centroid of ΔPQR and circumcentre of ΔABC is $\frac{5}{6}$

SECTION-3

Numerical Value Type Questions:

49. Let $f(x) = \log_{10}|2 + \sin x|$ then

$\int_{-1}^{1} \frac{\lim_{n\to\infty} (f(x))^n}{\sin^2 x + 3\sin x + 1} dx$ is equal to

50. Two perpendicular tangents are drawn to the ellipse $(x + y + 1)^2 + 2(x - y - 3)^2 = 4$ from a point. The chord of contact touches a circle concentric with the given ellipse. Then ratio of maximum to minimum radius of circle is

51. Let $f(x)$ be a differentiable function such that $f'(x) + f(x) = 4xe^{-x} \sin 2x$ and $f(0) = 0$.

If $\sum_{n=0}^{7} f\left(\frac{(4n+1)\pi}{4}\right) = e^{-\frac{\pi}{4}}\left(\frac{e^{-k\pi}-1}{e^{-\pi}-1}\right)$, then k is equal to

52. Let $A=\left[a_{ij}\right]_{n\times n}$ such that $a_{ij}=\dfrac{(-1)^i\left(2i^2+1\right)}{4j^4+1}$, then value of $1+\lim\limits_{n\to\infty}\left(\text{trace}\,A\right)$ is

53. If T_1, T_2, ..., T_{100} are the distinct terms in Harmonic progression and T_6, T_{10}, T_{18} are in Arithmetic progression. If T_p, T_{12}, T_q are in Geometric progression, then maximum value of q is
($p < 12 < q$)

54. Reflection of parabola $y^2 - 4x + 4 = 0$ in the line $x + y + 1 = 0$ is given by $x^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$, then $b + h + g + f + c$ equals

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